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Open Source Media Center (OSMC) releases Kodi 18.0 'Leia'

Kodi Version 18 adds support for retro-player gaming and associated game control. It also comes with audio and video playback improvements.

Kodi 18.0 Leia is available as a free download for all supported platforms, including Windows, Mac, Linux, iOS and Android.

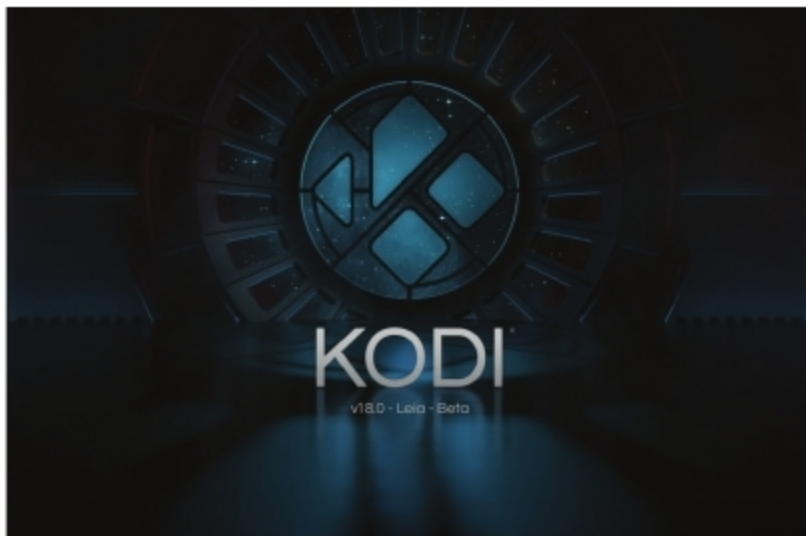
The new release also introduces support for joysticks, game pads, and other platform-specific controls for an immersive experience.

Kodi 18 comes with digital rights management decryption support. This means that depending on your hardware and licensing, Kodi can now access external DRM handlers and then play subscription content just like any other local media.

Other significant improvements include better filtering (media source, artist gender, type, artist name, enhanced artwork, etc), faster API access (particularly useful if you're controlling Kodi by remote with the TV off), and better sorting of your music library.

It also features improved Blu-ray support, audio and video playback improvements, and some modifications to the default skin, which is called Estuary.

"We've also improved Blu-ray support in terms of disc detection and metadata, and BD-J menu support (subject to Java support on the device). There



are updated external interfaces — for example, MPEG DASH and RTMP input, and there are improvements to 3D playback (including in 2D mode) and various changes to specific CODECs," the team at OSMC added.

On the audio side, there are a number of improvements and new support for all types of playback systems — ALSA, PulseAudio, OSS, Pi Audio, DirectSound, WASAPI, Darwin and SndIO.

Some of the changes will be invisible to many users – Bluetooth support, CMake build system, visualisations and screensavers, improvements to the JSON-RPC API, improved code stability, performance and security (as well as general code clean-up in many core areas), remote control changes, Web interface changes, logging changes, dependency changes, and many more.

The team has also designed the Kodi 18 shirt – which is available in several colours.

Open source mapping platform Mapzen joins the Linux Foundation

Mapzen, an open source mapping platform focused on the core components of map display, including search and navigation, is now a part of the Linux Foundation.

Facebook open sources DeepFocus for more realistic virtual reality images

Facebook has open sourced DeepFocus, an AI powered framework for improving the focus on close objects, to help others in the virtual reality (VR) research community.

DeepFocus works with advanced prototype headsets, rendering blur in real-time and at various focal distances.

The core technology behind DeepFocus is a novel end-to-end convolutional neural network that produces an accurate retinal blur as soon as the eye looks at different parts of a scene.

In a recent blog post, a member from the Facebook research team explained, "When someone wearing a DeepFocus-enabled headset looks at a nearby object, it immediately appears crisp and in focus, while background objects appear out of focus, just as they would in real life. This defocused blur (also known as retinal blur) is important for achieving realism and depth perception in VR."

The team also claimed that DeepFocus is the first system to be able to produce this effect in real-time for VR applications.

As research into new VR headset technologies advances, the Facebook team believes that DeepFocus will be able to simulate the accurate retinal blur needed to produce extremely lifelike visuals.

"The platform also demonstrates that AI can help solve the challenges of rendering highly compute-intensive visuals in VR. DeepFocus provides a foundation to overcome practical rendering and optimisation limitations for future novel display systems," the team wrote in the blog.

The team noted that as DeepFocus relies only on standard RGB-D colour and depth input, it can work with nearly all existing VR games and applications.

It is also compatible with all three types of accommodation-supporting headsets currently being explored in the VR research community: vari-focal displays (such as Half Dome), multi-focal displays and light field displays, the team members said.

The source code, network models and data sets for DeepFocus can be found in Facebook Research's GitHub repository.